

The decomposition of Philippine-type voice: implications for phasehood and movement¹

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main takeaway: Austronesian voice morphology is a composite (i.e. internally complex) phenomenon, and should be attributed to the spell-out of several distinct functional projections: Top, Mood, Appl

common syntactic analyses: voice morphology treated as a set of morphemes whose locus of spell-out is delegated to a single functional head (Rackowski and Richards 2005, Chen 2017, Erlewine et al. 2017)

- often in complementary distribution, as below...

(1) Seediq [Taiwan; Atayalic]

- a. ga=[ku] k<**mPROG=1SG.PIV <AV>cut chicken
'I am cutting chicken.' **actor voice (AV):** infix <m>**
- b. kret-**un**=mu [ka rudux]
cut-PV=1SG.GEN PIV chicken
'I will cut *the* chicken.' **patient voice (PV):** suffix -un
- c. s-keruc=mu rudux [ka yayu nii]
CV-cut=1SG.GEN chicken PIV knife this
'I will cut chicken with *this* knife.' **circumstantial voice (CV):** prefix s-

- NOTE: voice marker is **bolded**, pivot argument is in [brackets] and translated as *italics*

THIS TALK! there is no "voice" head, but an interaction of several functional categories...

- morphological and syntactic evidence converge to show the need for decomposition
- previous literature: decomposition in Malagasy (Guilfoyle et al. 1992, Paul 1999, Travis 2010), applicatives in Formosan (J. Wu 2007; H. Chang 2009, C.-K. Shih 2017, Bogren Svensson 2024)
- combines insights from theories on Austronesian voice as argument structure vs. information structure alternations (Schachter 1976, Aldridge 2004, Pearson 2005; R&R 2005; Chen 2017)

structural hierarchy: **Piv** > C > T > Neg > Asp > **Top** > **Mood** > Voice > **Appl** > v > V > √ROOT

- functional projection that is always present:
 - Mood: grammatical mood head, responsible for (ir)realis marking (cf. E, Travis 2010)
- functional projections that trigger argument advancement:
 - Top: object shift / topic head, at edge of extended Voice domain (R&R 2005, LaCerde 2020)
 - Appl: raising applicative head, reorders internal arguments (Georgala 2012)

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1 background - Austronesian and Philippine-type voice

Austronesian (*aka* symmetrical) voice systems (Dahl 1973, Wolff 1973, Schachter 1976, Blust 1977):

- morphological alternations on the verb reflect which argument—which can be core or non-core—is the **pivot**, a privileged argument in terms of word order, extraction, definiteness, case, and more
- further distinction between Philippine-type (2) and Indonesian-type voice
 - the former refers to a four-way distinction, present in Philippine/Taiwanese languages
 - the latter refers to a two-way (AV/PV) distinction, present in Western Austronesian languages

(2) Paiwan (Ferrell 1979) [Taiwan; Paiwan]

- | | | |
|----|---|----------------------------------|
| a. | q<m>alup [a cauc <u>au</u>] tua vavuy i gadu tua vulu | actor voice (AV) |
| | <AV>hunt PIV man OBL pig LOC mountain OBL spear | |
| b. | qalup- en nua cauc <u>au</u> [a vavuy] i gadu tua vulu | patient voice (PV) |
| | hunt-PV GEN man PIV pig LOC mountain OBL spear | |
| c. | qalup- an nua cauc <u>au</u> tua vavuy [a gadu] tua vulu | locative voice (LV) |
| | hunt-LV GEN man OBL pig PIV mountain OBL spear | |
| d. | si -qalup nua cauc <u>au</u> tua vavuy i gadu [a vulu] | circumstantial voice (CV) |
| | CV-hunt GEN man OBL pig LOC mountain PIV spear | |
| | 'The man hunts wild pigs in the mountains with the spear.' | |

- Seediq has a Philippine-type voice system, but with a three-way voice distinction
 - PV/LV are both used as undergoer voice, with a PFV/IPFV aspectual meaning (3) (Holmer 1996)

- | | | | | |
|-----|----|------------------------------------|----|----------------------------------|
| (3) | a. | wada=mu mah- un [ka sino] | b. | wada=mu mah- an [ka sino] |
| | | PST=1SG.GEN drink-PV PIV wine | | PST=1SG.GEN drink-LV PIV wine |
| | | 'I drank (and finished) the wine.' | | 'I was drinking the wine.' |

the **mapping between voice specification and pivot selection** is fairly consistent across languages

- descriptively, AV promotes the “subject”, PV promotes the “object”, and CV promotes the “adjunct”
- in trivalent constructions (ditransitive/causative/control), CV promotes the most embedded theme

	AV	PV	(LV)	CV
intransitive	<i>subject</i>	-	location	oblique
transitive	<i>agent</i>	<u>theme</u>	location	oblique
ditransitive	<i>agent</i>	goal	location	<u>theme</u>
causative	<i>causer</i>	causee	location	<u>theme</u>
control	<i>controller</i>	controllee	location	<u>theme</u>

there are a range of theoretical analyses of Austronesian voice...

- ergativity (de Guzman 1978; Aldridge 2004), active/passive voice (Wurmbrand & Shimamura 2017)
- case licensing (Guilfoyle et al. 1992), topic and $\bar{A}$ (Pearson 2005; van Urk 2015; Chen 2017)
- object shift (Rackowski & Richards 2005), phase extraction (Erlewine et al. 2017; Erlewine 2018)

why decompose? if we take morphology as the starting point...

- [ORIENTATION] different orientation (prefix vs. suffix): morpheme-specific or structural difference?
- [AFFIX ORDER] morpheme and word order reflects clause structure under the Mirror Principle: voice morphology is located in the middle field, not high in the left periphery (Travis 2010, Wu 2024)
- [IRREALIS] irrealis voice paradigm is morphologically complex with containment relation (Wu 2025)
- [CLAUSE SIZE] voice markers have different syntactic distributions with respect to adverbs and complementation, forming an implicational hierarchy (cf. ICH, Wurmbrand & Lohninger 2023)

2 morphological evidence – from Seediq and other Formosan languages

2.1 affix and auxiliary order (Wu 2024)

under Baker's (1985) Mirror Principle, *morpheme order and syntax should reflect one another...*

morpheme order with respect to root $\Leftrightarrow$ location within clause structure

vocabulary items: ²		head which hosts it:	
PFV	$\Leftrightarrow \langle n \rangle$	infix location __V	Asp
AV	$\Leftrightarrow \langle m \rangle$	infix location __V	Mood
REC	$\Leftrightarrow p\text{-RED}_{CV-}$		Voice[+D]
ANTIC	$\Leftrightarrow \emptyset_{V-}$		Voice[-D]
CV	$\Leftrightarrow s-$		Appl (to be revised)
CAUS	$\Leftrightarrow p-$		V_{CAUS}

- **actor voice (av) is located between Asp and Voice,³ in functional projection Mood**

– based on order of left-aligned affixes (cf. infixes are underlyingly prefixes, Kalin 2022)

#1: Asp > Mood perfective PFV attaches outside of AV and CV

- (4) a. Seediq
- | | | | |
|---|-------------|---|---------------|
| $\langle m \rangle$ osa | ‘AV>go’ | $r\langle m \rangle$ engo | ‘AV>talk’ |
| $\langle n \rangle$ osa | ‘PFV>go’ | $r\langle n \rangle$ engo | ‘PFV>talk’ |
| $\langle m \rangle \langle n \rangle$ osa | ‘PFV>AV>go’ | $r\langle m \rangle \langle n \rangle$ engo | ‘PFV>AV>talk’ |

- b. AV $\langle m \rangle$ and PFV $\langle n \rangle$ both target the same infix location __V:

osa ‘go’ $\rightarrow \langle m \rangle$ osa ‘AV>go’ $\rightarrow \langle m \rangle \langle n \rangle$ osa ‘PFV>AV>go’ $\rightarrow mnosa$
 rengo ‘talk’ $\rightarrow r\langle m \rangle$ engo ‘AV>talk’ $\rightarrow r\langle m \rangle \langle n \rangle$ engo ‘PFV>AV>talk’ $\rightarrow rmnengo$

- (5) a. Bunun (NTU 2016)
- | | | | |
|--|--------------|---|---------------|
| ma -ludah | ‘AV-hit’ | si -pit’ia | ‘CV-cook’ |
| l<math>\langle in \rangle</math> udah | ‘PFV>hit’ | p<math>\langle in \rangle</math> it’ia | ‘PFV>cook’ |
| m<math>\langle in \rangle</math> -ludah | ‘PFV>AV-hit’ | s<math>\langle in \rangle</math> -pit’ia | ‘PFV>CV-cook’ |

²Seediq orthography can be confusing due to pre-stress vowel lenition, where all vowels three or more syllables from the right edge are pronounced with [u], e.g. *tgdaya* [tugu-daya]. Thus, the prefixes (but not infixes, see Wu 2024) in the vocabulary items here have underlying vowels, though unspecified: REC $p\text{-RED}_{CV-}$ [pVCV-], ANTIC $\emptyset_{V-}$ [V-], CV $s-$ [sV-], and CAUS $p-$ [pV-].

³The outer Asp head is not to be confused with the inner Asp head (Travis 2010) which also present in Austronesian languages and is low in the clause (Voice > Asp_{RED} > Appl). This inner aspect is present in Seediq as well and spelled out via reduplication via the prefix RED_{CV-}, and like other Austronesian languages is indeed spelled-out closer to the root than AV morphology.

#2: Mood > Voice Av attaches outside of anticausative ANTIC and reciprocal REC

- (6) a. Seediq $d\langle m \rangle engu$ ‘⟨AV⟩dry’ $g\langle m \rangle aalu$ ‘⟨AV⟩care’
 $\emptyset_V-dengu$ ‘ANTIC-dry’ $p-g-gaalu$ ‘REC-care’
 $\langle m \rangle \emptyset_V-dengu$ ‘⟨AV⟩ANTIC-dry’ $\langle m \rangle-g-gaalu$ ‘⟨AV⟩REC-care’
- b. Av ⟨m⟩ targets the infix location __V and infixes *into* the ANTIC and REC affixes:²
dengu ‘dry’ → /V-dengu/ ‘ANTIC-dry’ → /⟨m⟩V-dengu/ ‘ANTIC-dry’ → *mtakur*
gaalu ‘care’ → /pV-gV-gaalu/ ‘REC-talk’ → /p⟨m⟩V-gV-gaalu/ ‘⟨AV⟩REC-talk’ → *mggaalu*
- (7) a. Puyuma $d\langle em \rangle isdis$ ‘⟨AV⟩tear’ $\langle m \rangle apit$ ‘⟨AV⟩pile.up’
 (Chen 2020) $u-disdis$ ‘ANTIC-tear’ $u-apit$ ‘ANTIC-pile.up’
 $\langle m \rangle u-disdis$ ‘⟨AV⟩ANTIC-tear’ $\langle m \rangle u-apit$ ‘⟨AV⟩ANTIC-pile.up’

- auxiliary and word order also suggests that **all verbal morphology is located below Asp**
 - auxiliaries: C [*netun* ‘if’] - T [*wada* ‘PST’, *maha* ‘FUT’] - Neg [*ini* ‘NEG’] - Asp [*ga* ‘PROG’]
 - these auxiliaries do not interfere with voice marking on verb in complex verb formation

#3: Asp > “voice” auxiliaries come before verbs that are marked AV/PV/LV/CV

- (8) a. $ga=[ku]$ $\langle m \rangle ekan$ ido d. $wada=[ku]$ ini $\emptyset$ -ekan
 PROG=1SG.PIV ⟨AV⟩eat rice PST=1SG.PIV NEG AV.IRR-eat
 ‘I am eating (rice).’ ‘I did not eat yesterday.’
- b. $ga=mu$ $puq-un$ [ka hei qhuni] e. $wada=mu$ ini $puq-i$ [ka hlama]
 PROG=1SG.GEN eat-PV PIV fruit PST=1SG.GEN NEG eat-PV.IRR PIV candy
 ‘I am eating *the fruit*.’ ‘I did not eat *the candy*.’
- c. $ga=mu$ $s-bege$ [ka patis] f. $wada=mu$ ini $biq-an-i$ [ka patis]
 PRPG=1SG.GEN CV-give PIV book PST=1SG.GEN NEG give-CV.IRR PIV book
 ‘I am giving (him) *the book*.’ ‘I did not give (him) *the book*.’

OBSERVATION #1: voice morphology is located in the middle field, *not* in the left periphery (i.e. C/T)

2.2 irrealis voice paradigm (Wu 2025)

voice morphology is *not* in complementary distribution in the irrealis paradigm...

- **irrealis voice forms** are used across imperative, negative, and negative imperative constructions
 - we can contrast the realis forms in (8a-c) to the irrealis (8d-f, 9) and hortative forms (10)

- (9) a. ekan ido!
 eat.AV.IRR rice
 ‘Eat (rice)!’
- b. $puq-i$ [ka ido]!
 eat-PV.IRR PIV rice
 ‘Eat *the rice*!’
- c. $biq-an-i$ heya [ka patis]!
 give-CV.IRR 3SG PIV book
 ‘Give him *the book*!’
- (10) a. $puq-a$ ido!
 eat-AV.HORT rice
 ‘Let’s eat (rice)!’
- b. $puq-a-i$ [ka ido]!
 eat-PV.HORT PIV rice
 ‘Let’s eat *the rice*!’
- c. $biq-an-a-i$ heya [ka patis]!
 give-CV.HORT 3SG PIV book
 ‘Let’s give him *the book*!’

- no overt “irrealis” marker—perhaps *-a* is HORT—instead varies with voice specification

- comparison of realis vs. irrealis paradigm (11) reveals that **voice forms are decomposable**

- #1 REAL forms appear to be in complementary distribution (except Puyuma, cf. Teng 2008)
- #2 IRR (and HORT) forms appear to be increasingly internally complex, from AV to PV/LV to CV

(11) realis vs. irrealis voice markers in Formosan (simplified)⁴

	REALIS				IRREALIS			
	AV	PV	LV	CV	AV	PV	LV	CV
Seediq:	⟨m⟩	-un	-an	s-	∅	-i	-i	-an-i
Paiwan:	⟨em⟩	-in	-an	si-	-u	-u	-i	-an
Bunun:	ma-	-un	-an	'is-	-a	-a-v	-a-v	'is- -a-v
Puyuma:	⟨em⟩	-a-u	-a-i	-an-a-i	∅	-u	-i	-an-i
Tsou:	⟨m⟩	-a	-i	-nen-i	-	-	-	-
Amis:	⟨om⟩	-en	-an	sa-	∅	-en	-	sa- -en

OBSERVATION #2: grammatical mood (REAL vs. IRR) features underlie the spell-out of voice morphology

if morphological complexity reflects structural complexity...

- **there is no single “voice” head**—so what functional categories does “voice” represent?
 - a containment relation in morphological *and* structural complexity: AV < PV/LV < CV
- functional projections: 1) AV has Mood, 2) PV has Top+Mood, and 3) CV has Top+Mood+Appl
 - these projections can be **spelled-out as portmanteaus vs. individual morphemes**
 - mediated by features on the Mood head, e.g. [REAL] vs. [IRR] vs. [HORT]

(12) Seediq

	REALIS	IRREALIS
AV:	Mood _[REAL] ⇔ ⟨m⟩	Mood _[IRR] ⇔ ∅
PV:	Top+Mood _[REAL] ⇔ -un/-an	Top ⇔ -i
CV:	Top+Mood _[REAL] +Appl ⇔ s-	Appl ⇔ -an

- variation in Formosan languages can be straightforwardly captured with vocabulary items

(13) Puyuma

Mood _[REAL] ⇔ ⟨em⟩	Top ⇔ -u/-i
Mood _[REAL] ⇔ -a / __ Top	Appl ⇔ -an
Mood _[IRR] ⇔ ∅	

(14) Paiwan

AV:	Mood _[REAL] ⇔ ⟨em⟩	Mood _[IRR] ⇔ -u
PV:	Top+Mood _[REAL] ⇔ -in/-an	Top+Mood _[IRR] ⇔ -i
CV:	Top+Mood _[REAL] +Appl ⇔ si-	Top+Mood _[IRR] +Appl ⇔ -an

OBSERVATION #3: voice morphology consists of up to three distinct morphemes, ROOT-Appl-Mood-Top

⁴Data on voice morphology is compiled from two sources: the 2018 grammar series on Formosan languages published by the Council of Indigenous Peoples, as well as the 2024 *Handbook on Formosan Languages* edited by Li, Zeitoun, & de Busser.

2.3 clause size and complementation (Wu 2026)

complement clauses have different degrees of morphological, syntactic, and semantic complexity...

distribution of voice morphology $\Leftrightarrow$ size/structure of complement clause

- **availability of voice and TAM marking** forms an implicational scale with different clause types
 - Implicational Complementation Hierarchy (ICH; Wurmbrand and Lohninger 2023)
- four levels of clause types (15), with increasing size and structure:
 - cf. gerunds in Tagalog (Hsieh 2020), restructuring in Austronesian (Wurmbrand 2016, a.o.)

#1	gerund	vP	bare verb, no voice or aspectual morphology; no embedded agent
#2	control	MoodP	AV-only, no non-AV or aspectual morphology; embedded agent is PRO
#3	tough	TopP	all voices possible, no aspectual morphology; embedded agent is PRO
#4	finite	CP	all voice and aspectual morphology possible; embedded agent is overt

- reflects position of voice/TAM projections in clause: $[_{CP} C Asp [_{TopP} Top [_{MoodP} Mood [_{vP} \sqrt{\quad}]]]]$

- (15)
- prading $[_{vP}$ phuma=nami paye npalic]
begin.AV plant=1PL.GEN rice glutinous
'Our planting of glutinous rice begins.' gerund: bare verb
 - prading=nami $[_{MoodP}$ <m>huma paye npalic]
begin.AV=1PL.PIV <AV>plant rice glutinous
'We will begin to plant glutinous rice.' control: AV verb
 - rriqu=mu $[_{TopP}$ slhay-an ka patis nii]
difficult.AV=1SG.GEN read-LV PIV book this
'It is difficult for me to read this book.' tough: non-AV verb
 - <m>kela=ku $[_{CP}$ <n>sa-an=na Awi ka Plsiya]
<AV>know=1SG.PIV <PFV>go-LV=3SG.GEN Awi PIV Puli
'I know that Awi went to Puli.' finite: finite verb

- availability of voice marking also corresponds to adverb hierarchy (Chang 2009; B. Svensson 2024)
 - event-oriented (all voices) < aspectual (AV and PV only) < speaker-oriented (bare adverb)

OBSERVATION #4: more structure corresponds to more availability of different kinds of voice morphology

2.4 summary

structural hierarchy: **Piv** > C > T > Neg > Asp > **Top** > **Mood** > Voice > **Appl** > v > V > $\sqrt{\text{ROOT}}$

#1 voice morphology is located in the middle field, within the domain of word formation Asp

#2 grammatical mood features (REAL vs. IRR) in Mood form the basis of voice morphology

#3 voice morphology is decomposable and involves three different heads: Top, Mood, and Appl

#4 a containment relation $AV < PV < CV$ holds in terms of morphological and structural complexity

in the next section, we will discuss the syntactic functions of the projections Top and Appl...

3 syntactic analysis

previous theories of voice alternations did not account for complex morphological behavior...

THIS TALK! making explicit the functional structure while maintaining existing insights in the literature

3.1 functional projections

within a Relational Grammar framework... Top and Appl represent argument advancement rules

(i) Mood is default (ii) Top is 2 over 1 (iii) Appl is 3 over 2

AV: Mood (1) PV: Mood and Top (2-to-1) CV: Mood and Appl and Top (3-to-2-to-1)

- argument advancement rules have to be stepwise (cf. Kinyarwanda, Kimenyi 1976, Bell 1983)

3.1.1 Mood

#1 Mood: responsible for **(ir)realis and/or (non)finite marking** (cf. E, Travis 2010)

- in Davidsonian event semantics, existential closure $\exists$ that binds the event argument e of verb
- cf. “default AV” in restructuring (16) (Wurmbrand 2016, Chang 2017) is akin to infinitival form

- (16) a. wada=[ku] c<(m)hungi <(m)ekan iyu
PST=1SG.PIV <AV>forget <AV>eat medicine
'I forgot to take medicine.' matrix subject is pivot: [V-AV V-AV]
- b. wada=mu chngiy-un <(m)ekan [ka iyu]
PST=1SG.GEN forget-PV <AV>eat PIV medicine
'I forgot to take the medicine.' embedded object is pivot: [V-pv V-AV]

- evidence from other languages also suggests a middle field position for Mood:
 - French: non-finite verbs appear after *pas* and before adverbs, finite verbs appear before both
 - literature on subjunctives (Rivero & Terzi 1995; Cinque 1999, Zanuttini et al. 2012, a.o.)

3.1.2 Top(ic) and Appl(licative)

these are syntactic projections that serve as **morphological reflexes of argument advancement...**

- cf. “leapfrogging” in object shift (Bobaljik 1995; Rackowski & Richards 2005)
- extracts arguments of different base positions, some are more structurally embedded

can be modeled as phase heads, with three phases in the clause structure...

(i) finite clause, PivP (ii) external arguments, TopP (iii) internal arguments, ApplP

[_{clause} **Piv** > C > T ... [_{ext. arg.} **Top** > Mood > Voice ... [_{int. arg.} **Appl** > v > V > $\sqrt{\text{ROOT}}$

- Top/Appl can be conceptualized as “outer” and “inner” passive, or “high” and “low” object shift
- activated *if* and *when* the pivot moves successively through to final position in left periphery

#2 Top(ic): middle topic position at edge of the extended Voice domain⁵ (LaCerde 2020)

- two places for information structure: 1) left periphery, 2) middle field
- evidence from other languages also suggests the existence of middle field topics:
 - Brazilian Portuguese: topic constituents can appear sentence initially or medially (17)
 - sentence-internal topics in Chinese (Paul 2002) and “low-IP” topics in Italian (Belletti 2004)

(17) Brazilian Portuguese (LaCerde 2020)

- a. [na festa de Natal]_{TOP} eu dei dois livros pra Maria
 in.the party of Christmas I gave two books to.the Mary
 ‘I gave two books to Mary at the Christmas party.’ left-periphery topic
- b. eu dei [na festa de Natal]_{TOP} dois livros pra Maria
 I gave in.the party of Christmas two books to.the Mary
 ‘I gave two books to Mary at the Christmas party.’ middle field topic

#3 Appl(icative): raising applicative head, **reorders internal arguments** (Georgala 2012)

- two types of Appl: 1) **thematic**—introduces arguments, 2) **expletive**—moves arguments
- cf. applicative analyses of cv (Rackowski 2002; Aldridge 2004; Chang 2009; Nie 2019)
- independent evidence: animate themes raise into Spec,Appl causing word order effects (18)

- (18) a. pi-imah=[ku] *begu laqi*
 CAUS.AV-drink=1SG.PIV soup child
 ‘I made the child drink soup.’ inanimate theme: [*theme* - **causee**]
- b. pe-emux=[ku] **Mahung laqi**
 CAUS.AV-hug=1SG.PIV Mahung child
 ‘I let Mahung hug the child.’ animate theme: [**causee** - *theme*]

- evidence from other languages also suggests the existence of internal movement:
 - (a)symmetric applicatives in Bantu (Kimenyi 1976; McGinnis 2001; Baker and Collins 2006)
 - literature on ditransitives (Larson 1988; Georgala 2012) and possessor raising (Deal 2013)

3.1.3 Piv(ot)

#4 Piv(ot): final landing site of pivots in the left periphery, available to long-distance extraction

- derives sentence-final pivot position (19) (cf. roll-up movement, Holmer 1996; Aldridge 2004)
- Piv encodes $\bar{A}$ -features such as [TOP] or [REL], separate from C which encodes [WH] or [Q]⁶

- (19) a. wada slluhe pure damac Seediq ckawas [ka Ape]
 PST learn.AV cook.AV food Seediq last.year PIV Ape
 ‘Ape learned to cook Seediq food last year.’
- b. ye m-kela wada sapah Pawan [ka Awi]?
 Q AV-know PST house Pawan PIV Awi
 ‘Does Awi know that Pawan went home?’

⁵This will be revised later on (in section 4) to simply be a phase extraction head, and $\bar{A}$ -features are only encoded in Piv.

⁶This is because question formation and pivot promotion are independent mechanisms (see section 4.2).

3.2 derivation

- pivot argument is merged in base location as a DP with topic feature [TOP]
- Piv/Top/Appl probe for DP target with [TOP] feature, and moves it into its specifier
- Top and Appl are projected when they create an **interpretational difference**
 - cf. Travis 1984 on Germanic V2 movement—subjects raise to Spec,TP but not Spec,CP
 - Top and Appl serve the function of rearranging which argument is the highest, otherwise their presence is vacuous, e.g. if Top were projected in an AV derivation, it would still attract DP_{AG}
 - probing for [TOP] feature circumvents locality (cf. van Urk 2015)
- the following illustrates AV (20), PV (21), and CV (22) derivations in a ditransitive construction

(20) AV construction:

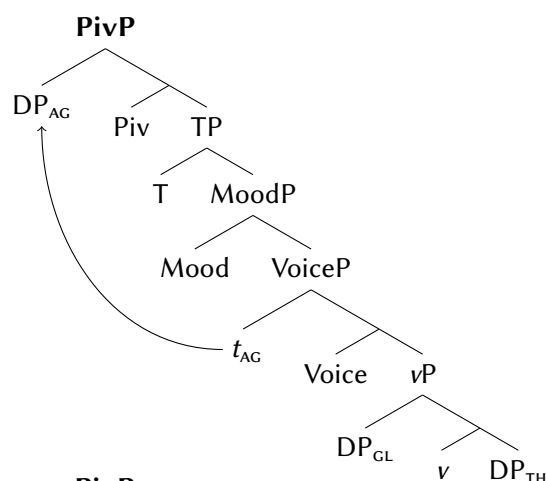
DP_{AG} has [TOP] feature

Piv finds DP_{AG}

→ DP_{AG} moves to Spec,PivP

Mood_[REAL] ⇔ ⟨*m*⟩

Mood_[IRR] ⇔ ∅



(21) PV construction:

DP_{GL} has [TOP] feature

Top finds DP_{GL}

→ DP_{GL} moves to Spec,TopP

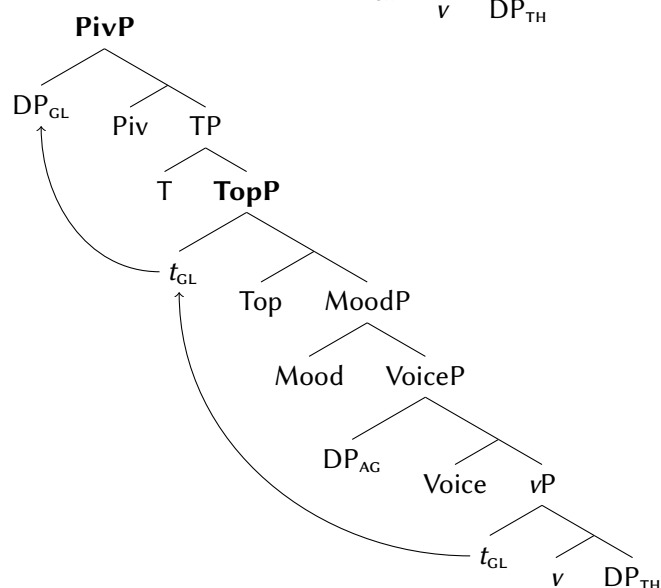
Piv finds DP_{GL}

→ DP_{GL} moves to Spec,PivP

Top+Mood_[REAL] ⇔ *-un/-an*

Top ⇔ *-i*

Mood_[IRR] ⇔ ∅



a note on movement mechanisms...

- there seems to be (?) no good way in our current frameworks to capture stepwise movement
 - appears violate locality by moving not the *highest* but the *second highest* argument
 - *but*, it does not allow unrestrained non-local movement (e.g. from base position to Spec,PivP)
- this is an issue for any movement motivated by phases, i.e. object shift, successive cyclic movement

(22) **cv construction:**

DP_{TH} has [TOP] feature

Appl finds DP_{TH}

→ DP_{TH} moves to Spec,ApplP

Top finds DP_{TH}

→ DP_{TH} moves to Spec,TopP

Piv finds DP_{TH}

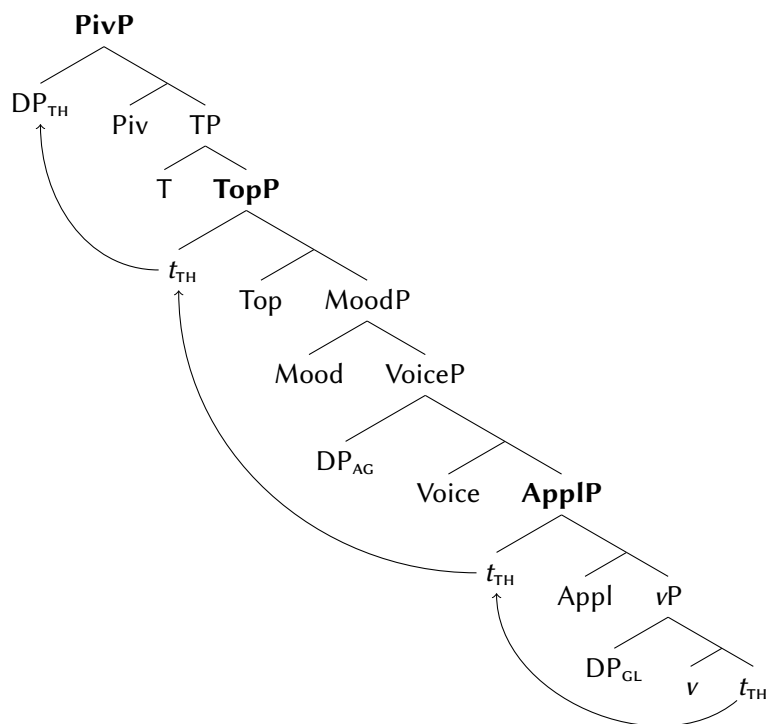
→ DP_{GL} moves to Spec,PivP

Top+Mood_[REAL]+Appl ⇔ *s-*

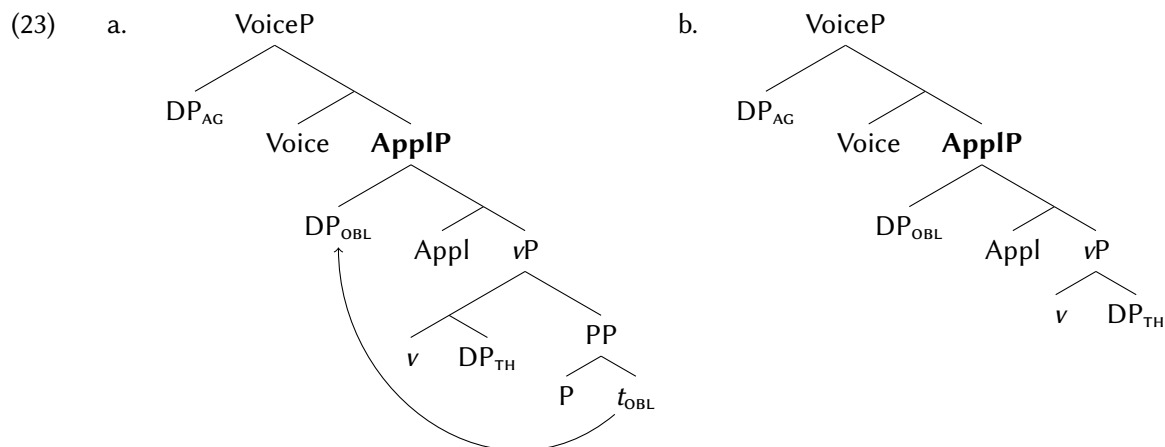
Top ⇔ *-i*

Mood_[IRR] ⇔ ∅

Appl ⇔ *-an*



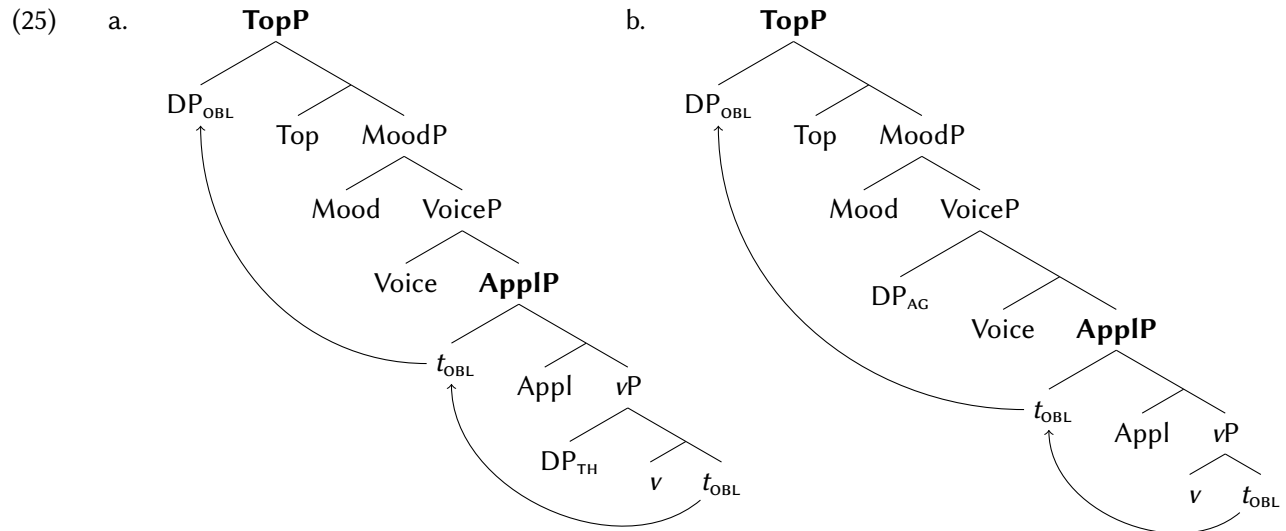
- two potential structures that can account for derivations with oblique arguments (23)
 - oblique argument merged in PP, raises to ApplP via preposition incorporation (van Urk 2015)
 - oblique argument directly merged in ApplP (Rackowski 2002; Aldridge 2004; R&R 2005)
- this is a matter of cross-linguistic variation, as evidenced by different binding relations (24)



- (24) a. *s-beebu kn-kingal laqi* [ka qreti=daha_i]
 cv-hit every child PIV stick=3SG.GEN
 'I hit every child_i with *their_i* stick.'
 TH > OBL
 (Seediq; Chen 2017)
- b. **i-b(in)antay ko ng bawat anak_i* [ang kanya=ng_i magulang]
 <PFV>cv-watch 1SG GEN every child PIV 3SG=LK parent
 'I watched every child_i for *their_i* parent.'
 *TH > OBL
 (Tagalog; Rackowski 2002)

- however, raising applicatives still needed in all languages for causative and control constructions

- potential issue: if Top and Appl are only projected to create an interpretational difference...
 - ...*but*, Top is vacuous in unaccusatives (25a) and Appl is vacuous in unergatives (25b)
 - tentative solution: intransitives are underlyingly transitive (?) (Hale and Keyser 1993)



3.3 additional evidence

- evidence to support that voice alternations are driven by $\bar{A}$ -movement, not A-movement

#1 pivot promotion is independent of nominative-accusative case assignment (Chen 2017)

- non-pivot intransitive subjects and transitive agents receive the same case value (26)⁷
- this would be unexpected under an ergative view of voice (de Guzman 1978, Aldridge 2004)

- (26) a. p-pyas-**an** na laqi [ka sapah=mu]
 RED-sing-LV GEN child PIV house=1SG.GEN
 ‘The children will sing in my house.’ (Chen 2023) unergative agent: GEN na
- b. h-hqil-**an** na riso nii [ka Paran]
 RED-die-LV GEN young.man this PIV Paran
 ‘This young man will die in Paran.’ (Chen 2023) unaccusative theme: GEN na

#2 pivots are topical in nature (Schachter 1976; Chen 2017)

- do not pass subject diagnostics (Dixon 1979) such as licensing imperatives or reflexives

- (27) a. qta-i [ka itu]!
 see-PV.IRR PIV that
 ‘Look at *that*!’
- b. puq-i [ka ido]!
 eat-PV.IRR PIV rice
 ‘Eat *the rice*!’
- (28) a. qta-un=na Pawan_i [ka **heya nanaq_i**]
 see-PV=3SG.GEN Pawan PIV 3SG REFL
 ‘Pawan_i sees *himself_i*.’
- b. *qta-un=na **heya nanaq_i** [ka Pawan_i]
 see-PV=3SG.GEN 3SG REFL PIV Pawan
 ‘*Himself_i* sees Pawan_i.’

⁷This case is marked GEN and not NOM because 1) it is syncretic with (and arose historically from) the 3sg possessive marker, and 2) in order to avoid confusion with ergative analyses of Austronesian voice systems that gloss the pivot marker as NOM.

4 implications — phasehood and movement

several parts of this analysis need refining...

[clause **Piv** > C > T ... [ext. arg. **Top** > Mood > Voice ... [int. arg. **Appl** > *v* > V > $\sqrt{\text{ROOT}}$

4.1 why phases?

I have posited Piv/Top/Apl as phase heads with the *sole function* of pivot promotion

- most transparently captures morphological behavior across Formosan languages
 - *if we accept the assumption that morphological complexity reflects structural complexity*, there does seem to be evidence that all these functional projections are present in that order
 - why phase heads in particular?
 - to derive **extraction restriction in Austronesian languages** (Keenan and Comrie 1977)
 - argument undergoing relativization must be reflected in voice morphology (29)
 - but see exceptions where non-pivot agents can be extracted (Hsieh 2020; Erlewine & Lim 2023)
- (29)
- | | |
|---|---|
| a. *patis [m]<n>atis (na/ka) Mahung
book <PFV><AV>write GEN/PIV Mahung | c. *sino [wada m]ege yaku (na/ka) Awi
wine PST <AV>give 1SG GEN/PIV Awi |
| b. patis [p<n>tas- an =na Mahung]
book <PFV>write-LV=3SG.GEN Mahung
'the book that Mahung wrote' | d. sino [wada=na s -bege yaku na Awi]
wine PST=3SG.GEN CV-give 1SG GEN Awi
'the wine that Awi gave me' |
- to motivate movement that corresponds to the various phase edges cross-linguistically
 - e.g. object shift, middle topics, successive cyclic movement, raising applicatives, etc.
 - arguments move to phase edges to **receive interpretations not available in base position**

this leads to several questions about locality and phasehood in movement...

1. what reasons are there—aside from phases—for this movement to be so local?
 - if we allow features like [TOP] or [REL], we don't run into the issue of Relativized Minimality
2. does attributing the phase function to Top have consequences?
 - theories on object shift posit Voice as phase head (cf. *v*, Chomsky 2001), such that 1) external and shifted arguments are both accessible and 2) this movement does not violate locality
 - however, some “trick” is still needed to get the internal argument *over* the external argument, whether that is with double specifiers, edge features, uninterpretable feature checking, etc.
3. is the concept of phasehood a proxy for other underlying mechanisms?
 - particular interpretations may only be available in certain projections or configurations

4.2 $\bar{A}$ -movement

we need to clarify the features and/or mechanisms that motivate pivot promotion...

- **voice morphology is employed for two functions: 1) pivot promotion, 2) relativization**

- so either [TOP] or [REL] can drive movement—or broadly [$\bar{A}$] features
- topics create new binding relations (30) (Chierchia 2024) but relativization does not (31)

- (30) a. *wada **m**-emux kn-kingal laqi_i [ka bubu=daha_i] *_{TH} > AG; WCO violated
 PST AV-hug every child PIV mother=3PL.GEN
- b. wada mex-**un** na bubu=daha_i [ka kn-kingal laqi_i] TH > AG; WCO obviated
 PST hug-PV GEN mother=3PL.GEN PIV every child
 ‘Their_i mother hugged every child_i.’
- (31) a. *kn-kingal laqi [wada **m**-emux (na/ka) bubu=daha_i] WCO violated
 every child PST AV-hug GEN/PIV mother=3PL.GEN
- b. *kn-kingal laqi [wada mex-**un** na bubu=daha_i] WCO violated
 every child PST hug-PV GEN mother=3PL.GEN
 intended: ‘every child_i that their mother_i hugged’

- question formation involves *wh*-in-situ (32) or predicate clefting (33) (Massam 2001)

- *wh*-words are prohibited in pivot position (32a), except in narrow contexts of echo questions
- Piv encodes $\bar{A}$ -features such as [TOP] or [REL], separate from C which encodes [WH] or [Q]

- (32) a. *wada <**m**>ekan hlama [ka ima]? (33) a. ima [ka wada <**m**>ekan hlama]?
 PST <AV>eat candy PIV who
 ‘Who ate candy?’ (ok: echo question)
 b. wada puq-**un** ima [ka hlama]? b. maanu [ka wada puq-**un**=su]?
 PST eat-PV who PIV candy
 ‘Who ate the candy?’ what PIV PST eat-PV=2SG.GEN
 ‘(The thing) that you ate is what?’

- where do these features occur in order to **derive movement through each phase edge?**

- option 1: all phase heads—Piv, Top, Appl—probe for DPs with [$\bar{A}$] features
- option 2: only Piv probes for [$\bar{A}$] features, other movement through Top and Appl is successive

it helps to distinguish “voice morphology” and “pivot promotion” as related but not equivalent concepts...

- **Piv (semantics) vs. Top/Mood/AppI (morphology):** have separate roles in voice systems

- semantic interpretation of [TOP] or [REL] on arguments—calculated at Piv
- surface realization of voice morphology on the verb—calculated at Top/Mood/AppI

- we should rename Top to reflect its role as a more general phase extraction head... any ideas?

- **NOTE:** there is likely a more nuanced view needed of the syntax-pragmatics interface

- information structure does not come from specific functional heads, but from relative positions and mapping rules in larger discourse templates (Neeleman and van de Koot 2006)
- allows interpretations to arise from a variety of positions, i.e. left periphery *and* middle field

- this cannot be captured by a featured-based system of [A]/[$\bar{A}$], although used here for exposition

4.3 inner and outer phase

is there cross-linguistic evidence for two phrases—*Top* and *Appl*—in the argument structure domain?

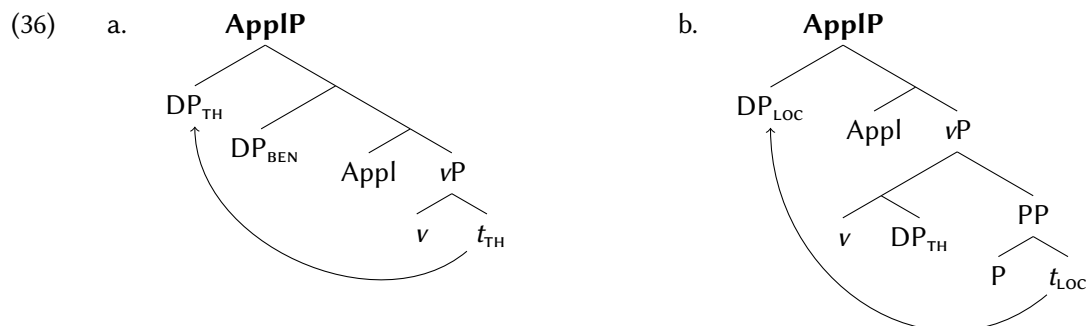
- *vP* as a phase (Chomsky 2001) has been used in two senses... maybe both are true!
 1. **internal to the argument structure domain** (*v* is distinct from Voice)
 2. **at the edge of the argument structure domain** (*v* is equivalent to Voice)
- evidence from cyclic phonology and morphology supports an inner phase (Travis 2010)
 - phonological processes that occur within vs. across phase boundaries
 - deletion of consonants in Malayalam (34) and vowels in Seediq² (35) within the inner phase

(34) Malayalam (Marantz 2007) [India; Dravidian]

- a. kuṭṭi booṭṭə mu-kk-i < [muŋŋ + -ikk]
 child boat sink-CAUS-PST
 ‘The child sank the boat.’
 lexical causative: [DP_{AG} [v_{CAUS} √ROOT DP_{TH}]]
- b. kuṭṭi booṭṭə muŋŋ-icc-u < [muŋŋ] + -ikk
 child boat sink-CAUS-PST
 ‘The child caused the boat to sink.’
 syntactic causative: [DP_{AG} v_{CAUS} [√ROOT DP_{TH}]]

- (35) a. *posa* p-osa < [pV- + osa] ‘CAUS-go’
pbeebu pu-beebu < [pV- + beebu] ‘CAUS-hit’
 b. *soosa* soʔ-osa < sV- + [osa] ‘cv-go’
sbeebu su-beebu < sV- + [beebu] ‘cv-hit’

- complements of phase heads are sent to spell-out, so *Appl* is not part of the inner cycle
- evidence from (a)symmetric applicatives in Bantu also supports an inner phase (McGinnis 2001)
 - oblique arguments must first be applicativized in order to be passivized, and in these passives benefactive constructions can also extract the theme, but locative constructions cannot
 - can be captured by *Appl* phase with variation in obliques merged in *Appl* vs. *PP* (36)



- a slight modification to proposals that use HighAppl (phase) vs. LowAppl (not a phase)
- the question is whether there is **evidence of both an inner and outer phase at the same time?**
 - is there one phase that can shift its location? or two phases with a set location?
- we may also have to revise our derivation such that *Top* and *Appl* are always projected

5 conclusion

main takeaway: Austronesian voice morphology is a composite (i.e. internally complex) phenomenon, and should be attributed to the spell-out of several distinct functional projections: Top, Mood, Appl

- *understanding morphology leads us to reevaluate our syntactic analyses of Austronesian voice*
- there are three phases—pivot moves through them successively to final position in left periphery

[_{clause} **Piv** > C > T ... [_{ext. arg.} **Top** > Mood > Voice ... [_{int. arg.} **Appl** > *v* > V > √ROOT

- greater articulation of functional categories within the argument structure domain
- separate roles of projections in voice systems: Piv (semantics) vs. Top/Mood/Apl (morphology)
- open questions:
 - can we reconcile this model of phrasal movement with more standard models?
 - how to derive “leapfrogging”—moving not the *highest* but the *second highest* argument?
 - does to extend this theory to Indonesian-type Austronesian voice systems?

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